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Correlating the mechanism, kinetics, and SEI formation of a boron-doped graphene anode for high-performance alkali ion batteries†

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Graphite has been used extensively for Li-ion batteries (LIBs) due to its low cost, abundant resources, and long-term stability. However, graphite exhibits limited capacity in LIBs, significantly poor capacity in Na-ion batteries (SIBs), and inferior cycling stability in K-ion batteries (PIBs), which hinders the development of sustainable next-generation battery technologies. Heteroatom doping and tuning interlayer spacing are known to be effective solutions for carbon-based materials. Such modified anodes can exhibit high capacity and cyclability with numerous active sites. Here, we report boron-doped thermally exfoliated graphene (BTEG), which exhibits a high reversible capacity of 1014 mA h g⁻¹ (LIBs), 295 mA h g⁻¹ (SIBs), and 369 mA h g⁻¹ (PIBs) at a current density of 25 mA g⁻¹. We used *operando/in situ* measurements such as Raman spectroscopy and electrochemical impedance spectroscopy to investigate the fundamentals involved during the real-time phenomenon of BTEG/alkali-ion batteries. Consequently, it helps assess the reaction mechanism and kinetics to establish a structure–performance relationship for all three batteries with the BTEG anode. Finally, based on solid electrolyte interphase (SEI) understanding and electrolyte tuning, highly reversible batteries were attained for up to 1000 cycles at a high current density of 1 A g⁻¹.

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1. Introduction

Lithium-ion batteries (LIBs) have propelled battery technology to new heights and revolutionized the design and development of portable electronics and electric vehicles in an unprecedented manner.^{1,2} However, the limited availability of lithium, constituting just 0.0017 wt% of the Earth's crustal abundance, will lead to a “lithium crisis” in the near future due to the growing demand for these batteries.³ The scarcity of lithium has generated significant interest in exploring more earth-abundant elements, such as sodium (Na) and potassium (K), as substitutes for lithium. Nonetheless, the relatively high standard reduction potentials of sodium (−2.74 V) and potassium (−2.96 V) in comparison to lithium (−3.04 V) constrain the energy density of Na-ion and K-ion batteries (SIBs and PIBs).⁴ Na- and K-ion-based systems can expedite the deployment of renewable

resources and their integration into power grids for large-scale energy storage applications where abundance holds greater importance than energy density.⁴ It is prudent to develop materials that will cater to all three kinds of alkali-ion batteries to provide comprehensive technological solutions for sustainability.⁵

Developing anode materials with high specific capacity plays a pivotal role in achieving high energy density in alkali-ion batteries. The substantial increase in ionic size when transitioning from Li⁺ (0.76 Å) to Na⁺ (1.02 Å) to K⁺ (1.38 Å) not only results in sluggish reaction kinetics but also leads to significant volumetric expansion, causing structural instability and electrolytic decomposition.⁶ Carbon, the most economical and abundant element in nature, has been extensively employed in various forms as an anode material for alkali-ion batteries.⁷ Graphite electrodes, discovered by Yazami in 1980, represent state-of-the-art anode materials for LIBs.⁸ The layered structure of graphite, upon intercalation with Li ions, exhibits a theoretical capacity of 372 mA h g⁻¹, low working potential, low volume expansion (12%), and a high reversible capacity.⁸ However, unlike Li-ion batteries, the unfavourable intercalation thermodynamics (in the case of SIBs) and substantial volume expansion of up to 150% (in the case of PIBs) limit the use of graphite as an anode for Na- and K-ion batteries.⁹ Therefore, it is imperative to develop carbon materials with tailor-made

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properties capable of delivering high energy density for LIBs, SIBs, and PIBs.⁵

In this regard, amorphous carbons, both in the form of hard (non-graphitizable) and soft (graphitizable) carbons, have demonstrated higher capacities and cycling stability for Li/Na/K-ion batteries compared to graphite, owing to their rich defects, porosity, and ability to accommodate significant volume changes.¹⁰ Hard carbons exhibit higher capacity compared to graphite. However, it is worth noting that a substantial portion of their capacity originates from a low, flat potential charge storage plateau (<0.2 V), which is the primary cause of sodium plating at fast cycling rates and, consequently, thermal runaway.¹¹ On the other hand, soft carbons, such as few-layered graphene nanosheets and carbons derived from coal tar, offer alternative solutions by providing capacity from sloping charge storage.^{12,13} Herein, carbon materials provide additional adsorption sites for $\text{Li}^+/\text{Na}^+/\text{K}^+$ due to their flexible pore size and high levels of surface defects.¹⁴ Additionally, the introduction of heteroatoms such as nitrogen (N), phosphorus (P), boron (B), and sulfur (S) further enhances the charge storage capacity by modifying the electronic properties of carbon, increasing the number of defects and creating more favorable sites for redox reactions.¹⁵ Heteroatom doping enhances the binding sites for alkali ions on the carbon surface and facilitates electron transfer, resulting in a significantly improved storage capacity.

Among various heteroatoms, boron has gained significant attention as it can modify the electronic and structural properties of carbon materials, influencing their performance in various applications.¹⁶ Boron creates an electron-rich environment for the surrounding carbon atoms due to its relatively lower electronegativity, enhancing Li^+ storage compared to unsubstituted carbon.¹⁷ Theoretical studies suggest that $\text{BC}_x\text{O}_{3-x}$ ($x = 1, 2, 3$) increases Li^+ ion intercalation compared to unsubstituted carbon.^{18,19} In the case of SIBs, boron-doped carbon containing BC_3 and B–C–O species can promote Na^+ ion movement due to reduced diffusion energy barriers.²⁰ Density functional theory (DFT) calculations showed that B, O co-doping can enhance potassium storage performance due to enhanced K^+ ion capturing capability by oxygen and enhanced conductivity by the B source. On the other hand, experimental studies on boron-doped carbon solely focus on the performance of lithium-ion batteries.^{21–23} The utilization of B doped carbon as a universal anode material for alkali-ion batteries holds significant potential for applications in portable electronics and electric vehicles; however, this approach remains relatively underexplored in the literature.

In this work, we report, for the first time, boron-doped thermally exfoliated graphene (BTEG) as a universal anode, demonstrating an overall enhanced performance for all three alkali-ion batteries. The as-prepared BTEG exhibits a high reversible capacity of 1014 mA h g^{-1} for LIBs, 295 mA h g^{-1} for SIBs, and 369 mA h g^{-1} for PIBs at a current density of 25 mA g^{-1} . The electrokinetic and mechanistic analysis from Dunn's equation, *operando* Raman spectroscopy, and the galvanostatic intermittent titration technique (GITT) indicate that the charge storage process for LIBs follows the intercalation-

adsorption mechanism. In contrast, SIBs and PIBs follow the dominant adsorption mechanism. The individual mechanisms involved in charge storage contribute significantly to the overall high performance of BTEG for alkali-ion batteries. We have also applied *in situ* electrochemical impedance spectroscopy (EIS) to probe diffusion dynamics and stability of BTEG with alkali-ion batteries in a half-cell configuration. In further contexts, we have addressed the issue of poor cycling stability for PIBs by understanding the nature of the solid electrolyte interphase (SEI) using *ex situ* X-ray photoelectron spectroscopy (XPS) analysis. The XPS analysis reveals that the highly concentrated electrolyte (HCE), *i.e.*, potassium bis(fluorosulfonyl)imide salt (5 M KFSI), results in the formation of a robust SEI which is essential to attain a high reversibility with BTEG. The effective passivation of the BTEG surface helped achieve electrochemical stability for up to 1000 cycles with a high-capacity retention of 87% at a high current density of 1 A g^{-1} for PIBs.

2. Results and discussion

2.1 Physicochemical and microscopic characterization studies

The synthesis of BTEG involved annealing thermally exfoliated graphene in the presence of boric acid under an Ar atmosphere. TEG was selected as the control sample and prepared by annealing thermally exfoliated graphene (TEG-350) in the absence of boric acid under the Ar atmosphere (Experimental section). Physico-chemical comparisons between BTEG, TEG and TEG-350 are provided through Powder X-ray Diffraction (PXRD), Raman spectroscopy, X-ray Photoelectron Spectroscopy (XPS) and nitrogen adsorption analyses (Fig. 1 and S1†). The PXRD patterns of both TEG and BTEG show broad peaks corresponding to the (002) plane of graphite in the range of $23\text{--}26^\circ$, reflecting poor crystallinity (Fig. 1a). The corresponding inter-layer spacing values for both TEG and BTEG are 0.36 nm , larger than that of graphite (0.336 nm). Fig. 1b shows Raman spectra of BTEG, which is defect-rich with an I_D/I_G ratio of 0.95 , slightly higher than the value obtained for TEG ($I_D/I_G = 0.93$). Fourier transform infrared (FTIR) spectra of BTEG show peaks corresponding to B–O (781 cm^{-1} , 1321 cm^{-1}) and B–C (1100 cm^{-1}) vibrations (Fig. 1c).²⁴ X-ray photoelectron (XPS) analysis of BTEG in Fig. 1d confirms the presence of boron to the extent of $2.7 \text{ wt\%}$, which is in agreement with the value obtained from TGA analysis ($3.7 \text{ wt\%}$) (Fig. S2†). The deconvoluted B 1s peak (Fig. 1d) of BTEG depicts four substitutional boron species: BC_3 ($\sim 188 \text{ eV}$), BC_2O ($\sim 190 \text{ eV}$), BCO_2 ($\sim 191.5 \text{ eV}$), and B–OH (193.2 eV).^{25,26} The boron species mainly comprises BC_2O (41.1%) and BCO_2 (51.1%) (Fig. S3a†), which suggests that boron replaces the carbon atom at the edge or defect sites in the graphene lattice which is evident from the signature peaks observed for carbon–boron and oxygen–boron bonds in carbon and oxygen XPS spectra, respectively (Fig. S3b and c†). The weak signal for BC_3 species (4.4%) indicates that the replacement of carbon atoms by boron in the graphitic framework is insignificant. The N_2 adsorption study shows a BET (Brunauer–Emmett–Teller) specific surface area (Fig. 1e) of $203 \text{ m}^2 \text{ g}^{-1}$ and $129 \text{ m}^2 \text{ g}^{-1}$ for TEG and BTEG, respectively.

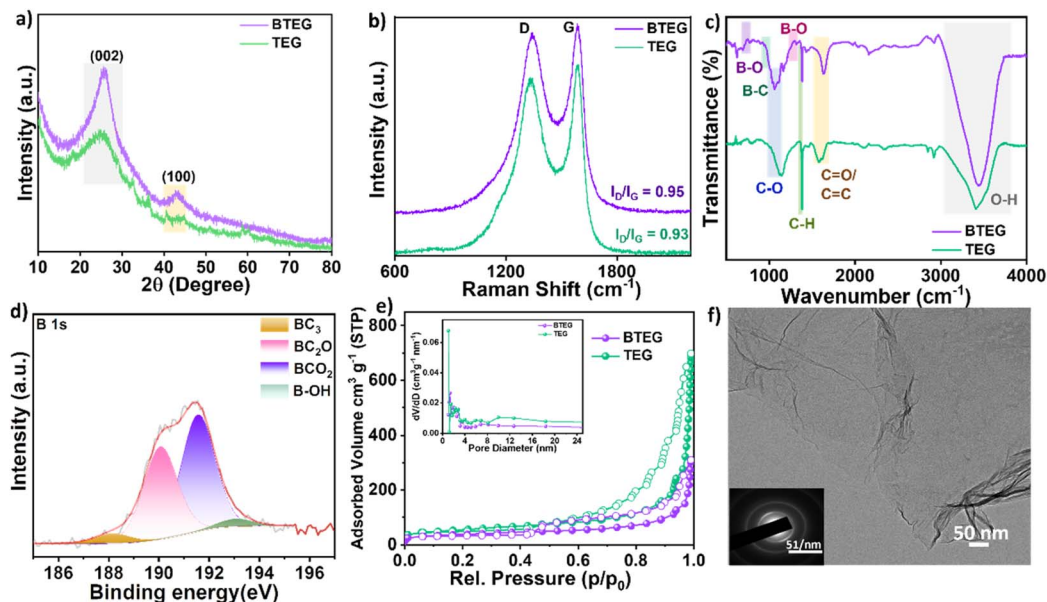


Fig. 1 (a) PXRD patterns, (b) Raman spectrum, (c) FT-IR of BTEG and TEG samples, and (d) X-ray photoelectron spectroscopy spectra of B 1s spectra. (e) N_2 adsorption/desorption isotherms of BTEG and TEG and (f) TEM image of BTEG with the SAED pattern (inset).

The transmission electron microscopy (TEM) image of BTEG (Fig. 1f) shows wrinkled ultrathin nanosheets similar to those of TEG (Fig. S4a[†]) which are also visible in the FESEM image in Fig. S4b.[†] The thickness of nanosheets measured from atomic force microscopy (AFM) (Fig. S4c[†]) was approximately 2.5 nm in the case of BTEG. Considering a typical graphene layer thickness of around 0.345 nm, the 2.5 nm thickness sheets correspond to seven graphene layers.²⁷ The selected area electron diffraction (SAED) pattern shows diffused diffraction rings for both TEG and BTEG (Fig. 1f inset).

2.2 Electrochemical characterisations for LIBs

The BTEG battery performance was studied using a half-cell configuration wherein the electrolyte chosen was 1 M $LiPF_6$ in ethylene carbonate and diethyl carbonate solvent (EC : DEC) for LIBs. Before conducting the electrochemical studies, BTEG assembled with Li metal was allowed to equilibrate for a 24-hour rest time (Fig. S5[†]). The open circuit voltage (OCV) for LIBs after the initial soaking period remains at 92% of the initial value.

The cyclic voltammetry (CV) technique was utilized to study electrochemical lithium redox behavior in BTEG. Fig. 2a shows the first five continuous CV curves within a voltage window of 3.0 to 0.01 V (V vs. Li/Li^+) at a low scan rate of 0.01 mV s^{-1} . In the first cycle, a prominent cathodic peak in the range of 1.2 to 0.6 V corresponds to the formation of the SEI from the decomposition of the electrolyte.²⁸ This peak disappears in the subsequent cycle due to SEI stabilization. The overlapping CV curves after the second cycle indicate a highly reversible interaction of lithium ions with boron-doped graphene layers of BTEG. The above results are consistent with the galvanostatic charge-discharge (GCD) curves of the BTEG electrode (Fig. 2b). BTEG shows an initial coulombic efficiency (ICE) of 51%, which is

higher than the value obtained for TEG (35%) (Fig. S6[†]). The low ICE of TEG can be due to its high surface area ($203 \text{ m}^2 \text{ g}^{-1}$), which leads to an increase in electrolyte decomposition and the irreversible capture of Li^+ ions.²⁹ After five cycles, BTEG and TEG exhibit a reversible charge storage capacity of 1014 mA h g^{-1} and 826 mA h g^{-1} at 25 mA g^{-1} , respectively, as shown in Fig. 2b and e. At a high current density of 1 A g^{-1} , BTEG shows an initial capacity of 276 mA h g^{-1} with a higher average coulombic efficiency of 99.7% than TEG (98.2%) (Fig. 2c). The capacity retention for BTEG is 75% after 1000 cycles of lithiation/delithiation processes. TEG occasionally shows an increase in capacity which could probably be explained by structural reorganisation leading to more access to storage sites.^{30–32}

To understand the reaction kinetics of Li^+ ions in BTEG before and after cycling (at current densities, 25 mA g^{-1} and 1 A g^{-1}), we recorded the EIS (Fig. 2d). Generally, there are three parts in the Nyquist plot, each associated with a different frequency range. The high-frequency region corresponds to bulk resistance (R_s) at high frequency, and medium frequency is associated with the solid electrolyte interface resistance (R_{SEI}) and the charge transfer resistance (R_{CT}).³³ Furthermore, the slope of the low-frequency part is due to the diffusional impedance/Warburg component (Z_W) of Li^+ ions within the BTEG electrode. We have not considered the electrolyte resistance component of R_s because it does not change much during battery aging. The R_{CT} value decreases from $107 \text{ } \Omega$ (in the charged state at open circuit voltage (OCV)) to $12 \text{ } \Omega$ after 10 cycles at 25 mA g^{-1} due to the activation and stabilization of the electrode by SEI formation.³⁴ Additionally, Z_W at OCV indicates capacitive-type behavior, likely due to electrolyte interaction with graphene layers, which changes to diffusive behavior after 10 cycles (stable SEI). The R_{CT} value of $17 \text{ } \Omega$ after 200 cycles at 1 A g^{-1} indicates robust behavior of BTEG and the SEI.

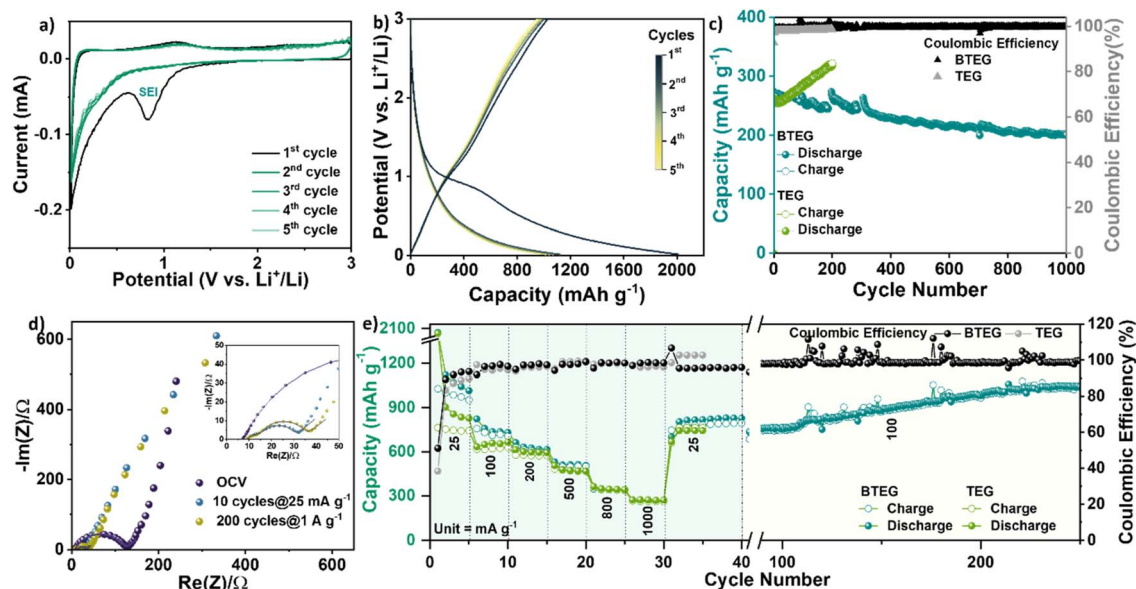


Fig. 2 The electrochemical properties of BTEG and TEG//Li half-cells. (a) CV curves at a scan rate of 0.01 mV s⁻¹ (0.01–3.0 V) of BTEG. (b) Charge–discharge curve of BTEG at 25 mA g⁻¹. (c) Long-term cycle stability at 1000 mA g⁻¹, (d) Nyquist plots of BTEG, and (e) BTEG and TEG rate capability at different current densities.

The rate performance analysis of BTEG shows a reversible capacity of 1014 mA h g⁻¹, 731 mA h g⁻¹, 615 mA h g⁻¹, 506 mA h g⁻¹, 342 mA h g⁻¹, and 261 mA h g⁻¹ at current densities of 25 mA g⁻¹, 100 mA g⁻¹, 200 mA g⁻¹, 500 mA g⁻¹, 800 mA g⁻¹, and 1000 mA g⁻¹, respectively (Fig. 2e). On the other hand, TEG shows a reversible capacity of 826 mA h g⁻¹, 665 mA h g⁻¹, 594 mA h g⁻¹, 484 mA h g⁻¹, 342 mA h g⁻¹, and 261 mA h g⁻¹ at current densities of 25 mA g⁻¹, 100 mA g⁻¹, 200 mA g⁻¹, 500 mA g⁻¹, 800 mA g⁻¹, and 1000 mA g⁻¹, respectively (Fig. 2e). Both BTEG and TEG show good retention (>80%) of their initial capacity when reverting to a lower current density of 25 mA g⁻¹. Long-term charge–discharge at 100 mA g⁻¹ was conducted with the BTEG electrode after the rate performance measurements. BTEG further shows an increase in capacity from 735 mA h g⁻¹ to 1042 mA h g⁻¹ at 100 mA g⁻¹, supporting its high charge storage capacity and stability. The increase in discharge capacity of BTEG from 100 to 250 cycles can be attributed to the activation process of the BTEG nanosheets. On continuous lithiation and delithiation, the nanosheets can structurally reorganise to enhance ion transport within active intercalation sites, which can increase the capacity over cycling.

Furthermore, commercial LiFePO₄ was used as the cathode and BTEG was used as an anode to show practical applicability of BTEG, as shown in Fig. S7 and S8.† The full cell performance analysis shows high coulombic efficiency and stable capacity retention which emphasise the material's structural robustness and electrochemical reversibility under practical conditions.

2.3 Electrochemical characterization for SIBs and PIBs

The performance of BTEG for SIBs and PIBs was studied using the electrolyte 1 M NaPF₆ in EC : DEC : FEC (10 vol%) and 5 M KFSI in EC : DEC, respectively. After equilibration for 24 hours

(Fig. S5†), the change in open circuit voltage (OCV) is 98% for SIBs and 99% for PIBs.

The BTEG sample shows an ICE of 27% and 29% for SIBs and PIBs, respectively, higher than TEG (10% and 16% for SIBs and PIBs, respectively) (Fig. S9a, b, S10a and b†). Furthermore, BTEG exhibits a reversible capacity of 295 mA h g⁻¹ for SIBs and 369 mA h g⁻¹ for PIBs at 25 mA g⁻¹ (Fig. S9a and b†). The relatively higher capacity value of BTEG for PIBs than for SIBs is due to highly concentrated 5 M KFSI (Fig. S10b†) salt over 1 M NaPF₆ (Fig. S9c†). In comparison, TEG exhibits only 83 mA h g⁻¹ and 225 mA h g⁻¹ for SIBs and PIBs, respectively (Fig. S10a and b†). Even at a high current density of 1 A g⁻¹, BTEG exhibits a capacity of 147 mA h g⁻¹ for SIBs, which is about seven times higher than that of TEG (21 mA h g⁻¹) (Fig. 3a). Similarly, BTEG shows a capacity value of 184 mA h g⁻¹ for PIBs, which is about two times higher than that of TEG (83 mA h g⁻¹), as shown in Fig. 3b. For TEG, the charge–discharge profile at 1 A g⁻¹ is given in Fig. S11.† This increment in capacity for BTEG in SIBs and PIBs is possibly due to enhanced surface defects associated with surface-based boron dopant species (BCO₂ and BC₂O species dominantly).^{19,20} In long-term cycling, BTEG shows positive capacity (>100%) and retains 87% of its initial capacity after 1000 cycles for SIBs and PIBs, respectively. Additionally, BTEG shows a high average coulombic efficiency of 99.9% and 99.95% for SIBs and PIBs, respectively. EIS studies of BTEG reveal a higher resistance for SIBs than for PIBs at OCV (after equilibration time).³⁵ This can be due to differences in the Na⁺/K⁺-solvation sphere, counter anions (PF₆⁻ and FSI⁻) and solvent of electrolyte (EC : DEC and EC : DEC : FEC). After 200 cycles at 1 A g⁻¹, a slight decrease in the R_{CT} value from 215 Ω to 170 Ω (10 cycles) for SIBs was observed, while a significant decrease in the R_{CT} value from 721 Ω to 332 Ω (10 cycles) for PIBs was observed (Fig. 3c and d). Therefore, after SEI stabilization the R_{CT} for PIBs is higher compared to SIBs

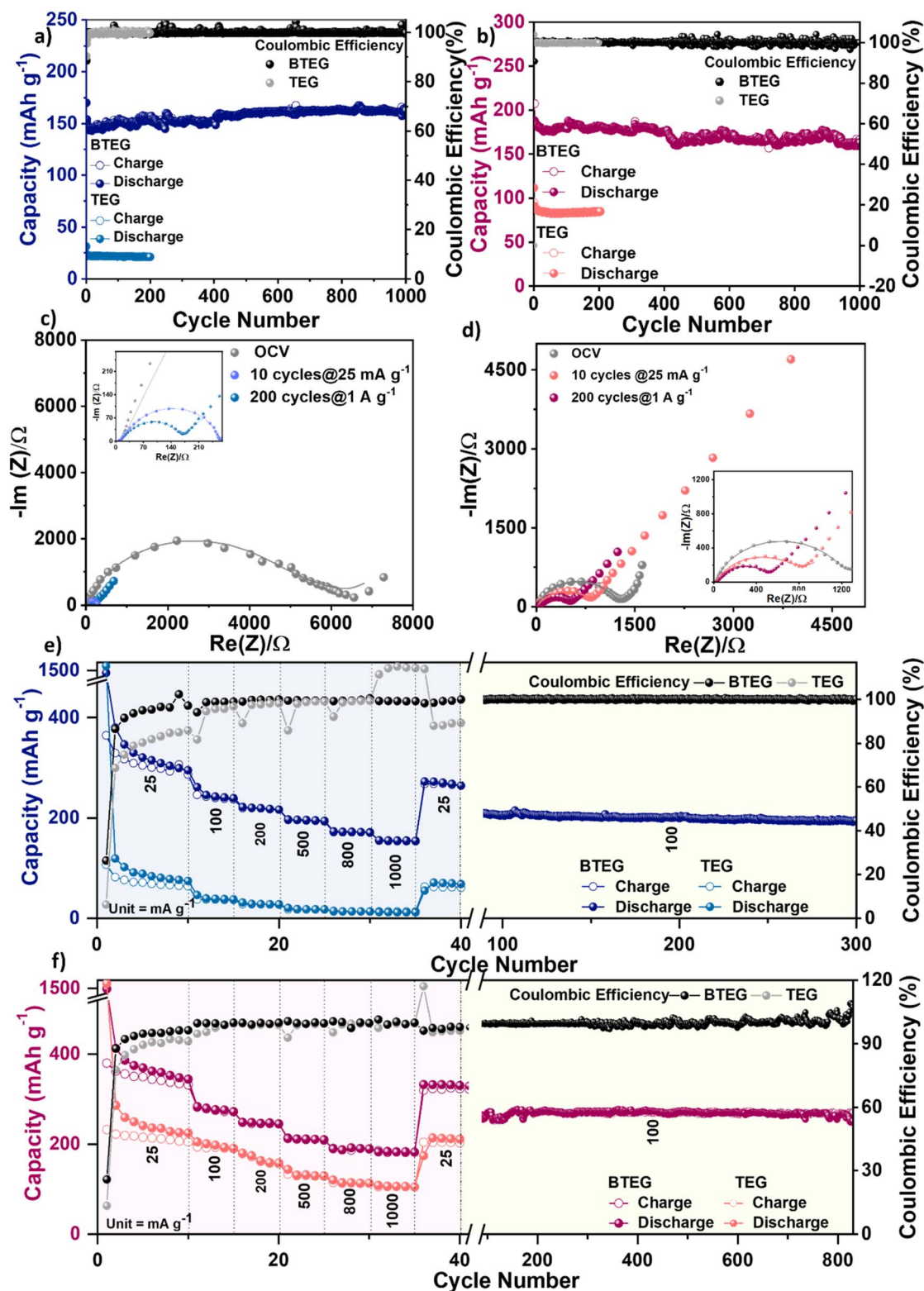


Fig. 3 The electrochemical properties of BTEG and TEG//Na half-cells. (a) Long-term cycle stability of BTEG and TEG at 1000 mA g⁻¹, (c) Nyquist plots of BTEG, and (e) rate capability of BTEG and TEG at different current densities. The electrochemical properties of BTEG and TEG//K half-cells. (b) Long-term cycle stability of BTEG and TEG at 1000 mA g⁻¹, (d) Nyquist plots of BTEG, and (f) rate capability of BTEG and TEG at different current densities.

(discussed in Section 2.4), and R_{CT} values decrease on cycling suggesting improved reaction kinetics for both SIBs and PIBs.

The reversible capacities at different current densities for TEG and BTEG with SIBs and PIBs are tabulated in Tables S1 and S2† respectively. BTEG demonstrates superior rate capability relative to TEG for SIBs and PIBs (Fig. 3e and f). BTEG shows 92% capacity retention on returning to 25 mA g⁻¹ after 40 cycles and 95% capacity retention after 300 cycles at 100 mA g⁻¹ for SIBs. Similarly, BTEG shows 96% capacity retention after 40 cycles on returning to 25 mA g⁻¹ and 100% capacity retention after 840 cycles at 100 mA g⁻¹ for PIBs. Such a stable and reversible charge storage behavior for BTEG with PIBs can be attributed to the role of the electrolyte in forming a stable SEI (discussed in Section 2.5).³⁶

To enhance the initial coulombic efficiency (ICE), we employed an electrochemical pre-alkalization process. Without pre-lithiation, pre-sodiation, or pre-potassiation, the ICE values were 51%, 27%, and 29%, respectively. However, following the pre-alkalization process, all three systems exhibited ICE values exceeding 90%, as shown in Fig. S12.†

2.4 Electrochemical kinetic analysis for understanding the alkali-ion storage mechanism in BTEG

Electrokinetic studies were carried out to gain a deeper insight into the superior electrochemical characteristics of BTEG electrodes for

alkali ion storage. The cyclic voltammetry technique provides insights into the reversibility of a reaction and helps determine whether the charge storage mechanism is diffusion-controlled or capacitive-controlled. The reversibility of the electrochemical reaction can be measured by observing overlap in CV curves at a particular scan rate as shown in Fig. S13a, b, S14a and b.† The reaction kinetics was measured using CV curves at different scanning rates (0.01–0.7 mV s⁻¹) for alkali ion batteries (Fig. 4a–c). The reversibility of the reaction can be seen by overlapping CV curves at 0.1 mV s⁻¹ for 5 cycles. Peak current increases with an increase in the scan rate and is related using the equation:^{37–40}

$$i = a\nu^b \quad (1)$$

where a and b are constants and b could be estimated from the slope of the curve plotted between $\log(\nu)$ versus $\log(i)$. A value of 0.5 b indicates a diffusion process, while 1 corresponds to a surface-capacitive process. Fig. 4d shows a plot of $\log(i)$ versus $\log(\nu)$ with a linear relationship, $R^2 = 0.99$. In the case of LIBs, BTEG exhibits a pseudocapacitive process validated using the value to be 0.57 for the anodic peak and 0.92 for the cathodic peak.^{41,42} The values indicate a diffusion-controlled process in the anodic reaction and a diffusive-capacitive process in the cathodic reaction for LIBs. Similarly, the values calculated for

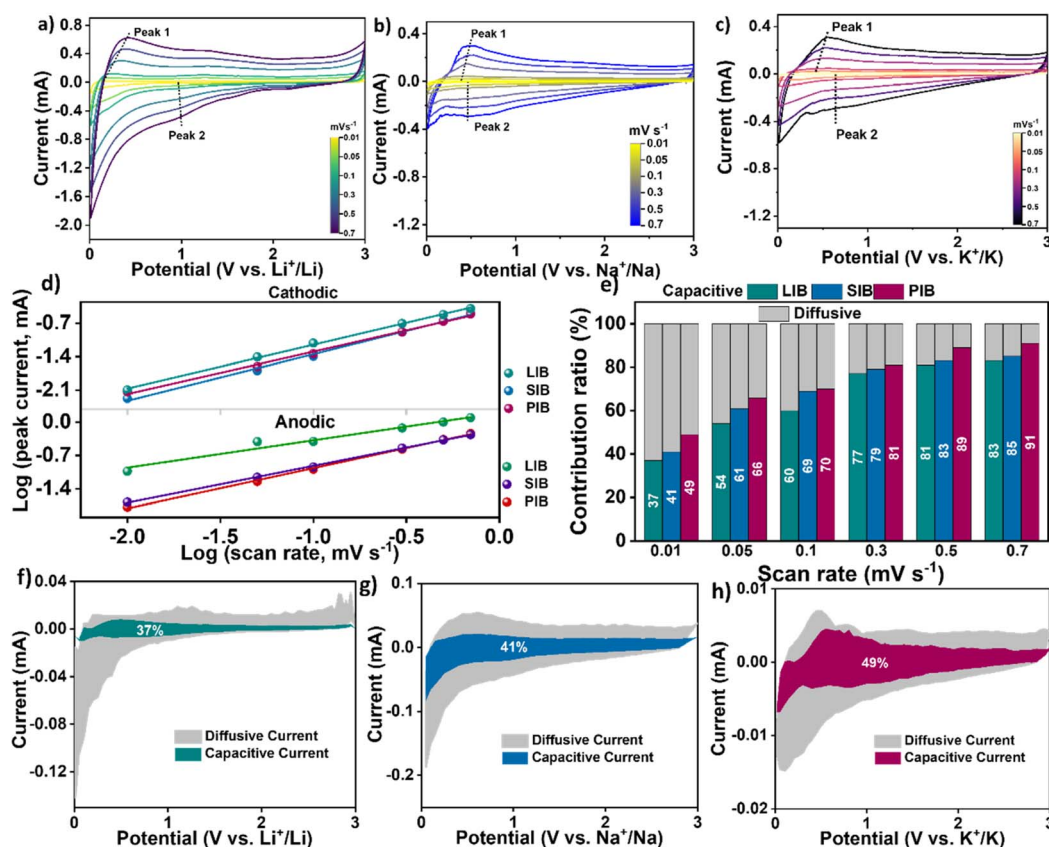


Fig. 4 Electrochemical kinetic analysis of BTEG (vs. M/M⁺). Cyclic voltammetry at different scan rates (0.01, 0.05, 0.1, 0.3, 0.5, and 0.7 mV s⁻¹) for (a) LIB (1 M LiPF₆ EC : DEC), (b) SIB (1 M NaPF₆ EC : DEC : FEC) and (c) PIB (5 M KFSI EC : DEC). (d) The correlations of the current scan rate (ν) and peak current (i) for anodic and cathodic processes compared for LIBs, SIBs, and PIBs. (e) Capacitive and diffusive (grey) contributions at various scanning rates for alkali ion batteries. The CV curve with shaded regions shows capacitive contribution for (f) LIB (cyan), (g) SIB (blue), and (h) PIB (wine) at 0.01 mV s⁻¹.

SIBs are 0.77 (anodic) and 0.98 (cathodic), and for PIBs, they are 0.84 (anodic) and 0.89 (cathodic). These high values suggest diffusive-capacitive processes for both SIBs and PIBs ascribed to the larger size of Na^+ and K^+ ions as compared to Li^+ ions.⁴³

Moreover, to quantitatively determine the contribution from the capacitive and diffusion-controlled processes, Dunn's equation⁴⁴ was used in CV curve analysis:

$$i = k_1 v + k_2 v^{1/2} \quad (2)$$

Here, v stands for the scan rate and k_1 and k_2 constants can be determined from the $i(V)/v^{1/2}$ versus $v^{1/2}$ plot. The $k_1 v$ and $k_2 v^{1/2}$ represent the capacitive and diffusion processes, respectively. Accordingly, at a scan rate of 0.01 mV s^{-1} , the capacitive-controlled process contributes 37% of the total current (charge-discharge) for LIBs, 41% for SIBs, and 49% for PIBs in the CV profile (Fig. 4e–h). The capacitive contribution increases with an increase in alkali-ion size, which can be attributed to an increase in preferable interaction with exposed surface-active sites for Na^+ and K^+ ions compared to dominant intercalation of Li^+ ions within BTEG nanosheets. When the scan rate increases to 0.7 mV s^{-1} , the contribution from the capacitive-controlled process enhances up to 83% for LIBs, 85% for SIBs, and 91% for PIBs (Fig. 4e). A pseudocapacitive mechanism indicated by the shaded region in the cyclic voltammogram is shown across all scan rates in Fig. S15–S17† for alkali ion batteries. The dominant non-faradaic contribution with increasing scan rates is the reason for extraordinary rate performance and can be attributed to the B–O/C–O functionalities in BTEG.⁴⁵

Furthermore, *operando* Raman studies were conducted to probe the charge storage sites during charge/discharge in BTEG for alkali ion batteries. The microstructural evolution was observed on discharging from 3 V (denoted as D3V) to 0.01 V (denoted as D0.01V) and recharging to 3 V (denoted as C3V). The G-peak corresponds to the first-order Raman scattering from the stretching of sp^2 atoms in crystalline graphene sheets (in-plane E_{2g} symmetry phonon).^{46,47} Meanwhile, the D-peak corresponds to disorder scattering, *i.e.*, in-plane sp^2 atoms being close to the K-point in the graphene Brillouin zone.⁴⁸ The G and D peak positions and intensities are affected by the degree of interlayer stacking and disordering of graphene, as well as electron transfer from intercalated species into graphene-based conduction bands. Therefore, charge transfer from Li/Na/K affects the D and G-peak energy, intensity and width.⁴⁶ In LIBs (Fig. 5a), upon lithiation (D3V), the pristine D and G band peaks start broadening below 1.25 V, indicating the interaction of Li^+ ions with active defect and graphitic sites. The G peak energy starts to downshift as intercalation progresses, corresponding to the electron doping of the graphene sheets. The electron doping at the intercalation site can be explained by the charge transfer from the Li atoms to π^* antibonding orbitals of C–C (graphene), which results in the lengthening of these bonds.⁴⁹ The D peak energy does not undergo shift, due to the fact that Li^+ ions adsorb onto the defects and B–C–O/C–O functionalities; thus, breathing motion of sp^2 atoms of graphene is limited by adsorbed Li^+ ions. Furthermore, variations in the intensity of the G and D peaks are influenced by shifts in

phonon energies and lifetimes, which result from electron-phonon interactions.^{50,51} The progressive effect on peak energy shifts and intensity indicates that the charge transfer delocalizes homogeneously across BTEG. Furthermore, at 0.01 V on complete lithiation, G and D peaks disappear due to loss of optical skin depth after highly conducting LiC_x is formed.⁵² The reverse trend, *i.e.*, a gradual increase in D and G peak intensity and energy, occurs on delithiation. LIBs support the intercalation-adsorption model, which is in agreement with the diffusive-capacitive process (Fig. 4f).

In the case of SIBs during discharge, the G peak experiences a redshift, yet peak intensity remains intact even at full discharge, while the D peak intensity diminishes drastically (Fig. 5b). On desodiation, D and G peaks are reversibly restored with the same peak position and width. Fig. 5c shows a similar trend in the case of PIBs, and the Raman results for SIBs and PIBs corroborate the capacitive controlled mechanism indicating the dominant adsorption model for SIBs and PIBs (Fig. 4g, h and 5b and c). This is also supported by the evidence that the I_D/I_G ratio for LIBs exhibits the disappearance of both D and G peaks where the G band shows early disappearance on lithiation. While SIBs and PIBs show a reduced I_D/I_G ratio, with a diminished D band and an intact G band on sodiation and potassiation (Fig. 5d), it is noteworthy that the I_D/I_G ratio for LIBs (initial: 0.88; final: 0.91), SIBs (initial: 0.85; final: 0.85), and PIBs (initial: 0.88; final: 0.83) are restored after cycling, showing high reversibility of BTEG.

Therefore, from the above observations and analysis, the reaction mechanism for the BTEG anode for alkali ion batteries can be clearly presented using a schematic illustration (Fig. S18†) and explained as follows. The Li^+ ion is known to readily intercalate to form a stable graphite intercalation compound (GIC) *i.e.*, LiC_6 due to its small ionic radius ($0.76 \text{ \AA}$) and can adsorb with surface defects.⁵³ Therefore, the BTEG anode exhibits an intercalation-adsorption model in LIBs. However, intercalation of Na^+ ions is inferior due to its larger ionic radius ($1.02 \text{ \AA}$) and thermodynamically unfavourable GICs (NaC_6 and NaC_8).⁵⁴ Additionally, in the case of PIBs, the intercalation of K^+ ions is inhibited within BTEG due to its larger ionic radius ($1.38 \text{ \AA}$), even though the K-GIC (KC_8) is thermodynamically stable. Therefore, the BTEG anode exhibits a dominant adsorption model in SIBs and PIBs.

The diffusion kinetics of the alkali metal ions on charge storage sites in BTEG were determined by the galvanostatic intermittent titration technique (GITT) with a pulse current of 25 mA g^{-1} (Fig. S19†). Fig. 5e and S20† show that the highest diffusion coefficient of alkali ions (D_{M^+}) is at 2.5 V which falls steeply at lower potential on discharging. This change in D_{M^+} is ascribed to rapid initial diffusion on vacant sites on BTEG initially, followed by a decrease in diffusion due to the repulsive charge gradient caused by the previously bound M^+ at the defect sites. D_{M^+} values above 1.25 V can correspond to alkali metal adsorption on the surface of boron-doped graphene layers. Below 1.25 V, the diffusion coefficient for Li^+ (D_{Li^+}) fluctuated from 10^{-11} to 10^{-10} , possibly due to staging-type charge storage/intercalation into graphitic layers.⁵⁵ Meanwhile, SIBs and PIBs show a smooth decrease in potential due to the interaction of

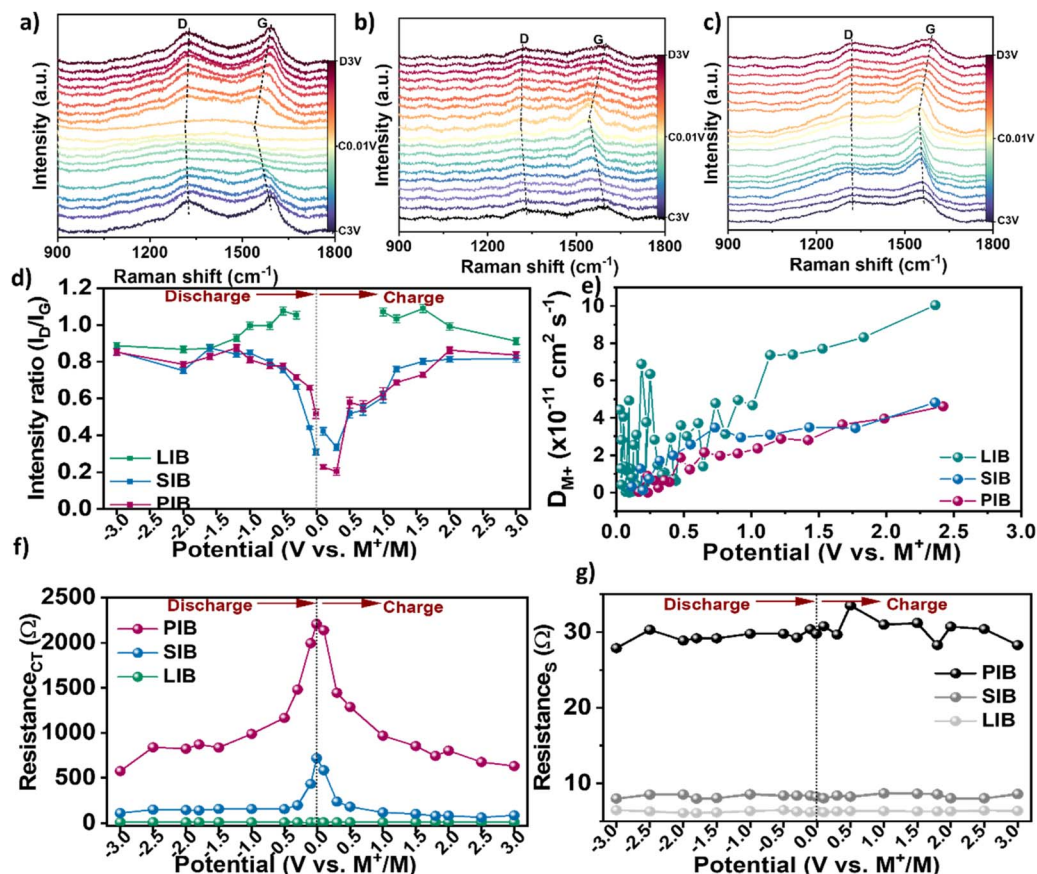


Fig. 5 Structural and kinetic evolution of BTEG with LIBs, SIBs and PIBs. *Operando* Raman spectra of (a) LIB, (b) SIB, and (c) PIB on discharging (3–0.01 V) to charging (0.01–3 V). (d) Evolution of the intensity ratio for a defect to the graphitic domain in BTEG with change in the potential for LIBs, SIBs, and PIBs. (e) The calculated chemical diffusion coefficients for LIBs, SIBs, and PIBs during the discharge process. *In situ* impedance spectra for LIBs, SIBs, and PIBs. Impedance associated with (f) charge transfer resistance and (g) solution resistance for LIBs, SIBs, and PIBs.

Na^+/K^+ with dopants and defect sites. LIBs exhibit a higher diffusion coefficient than SIBs and PIBs (10^{-11} to 5×10^{-11}) due to abundant intercalation-defect based coverage in BTEG with Li^+ ions, compared to only defect-based coverage (surface-dominant) for Na^+/K^+ ions (Fig. 5e) (as seen from Raman analysis, Fig. 5d).

In situ PEIS was used to analyze the reaction dynamics of both sides in a half cell, *i.e.*, BTEG (working electrode) and alkali metal (counter electrode), during the charge/discharge process. The EIS results are shown in Fig. 5f and g and obtained after SEI stabilization, *i.e.*, 10 cycles. Nyquist plots show that on discharging in BTEG, R_{CT} for BTEG/alkali metal, (obtained by fitting of *in situ* PEIS plots in Fig. S21† and using Fig. S22† as an equivalent circuit), increases with a decrease in potential due to increased coverage of charge storage sites corroborating with the GITT results (Fig. 5e and f). On fitting the results using an equivalent circuit as shown in Fig. S22†, it was observed that the R_{CT} values increase with increased alkali metal size. At the initial state (3 V), R_{CT} values increase by 11 times for SIBs and by 57 times for PIBs with respect to LIBs. On complete discharge (0.01 V), R_{CT} changed insignificantly for LIBs, 6.5 times for SIBs, and 3.8 times for PIBs compared to their initial state. The R_{CT} values confirm that LIBs exhibit superior and facile kinetics to SIBs and PIBs. Also, R_s for LIBs (6 Ω) and SIBs (8 Ω) are similar,

while PIBs (30 Ω) show a much higher value, which can be due to the use of HCE for PIBs (5 M KFSI) (Fig. 5g). There is an insignificant change in R_s and R_{SEI} for LIBs, SIBs, and PIBs on charging/discharging, showing good stability of electrolyte and the SEI with BTEG (Fig. 5g, S23 and S24†).

Additionally, previously observed resistance values for LIB, SIB, and PIB cycling data show facile charge transfer kinetics and a robust SEI, confirming long-term stability (Fig. 2d, 3c and d). Despite K experiencing higher resistance compared to Li and Na, BTEG maintains stability for 1000 cycles. Such high stability for PIBs, in particular, is attributed to the alteration of the electrolyte from the traditional 0.8 M KPF_6 to 5 M KFSI electrolyte. The significance of selecting the appropriate electrolyte and its optimisation for cycling stability is explored in the following section.

2.5 SEI on the BTEG surface with various electrolytes

The type of electrolyte, the nature of the SEI, and the alkali ion significantly influence the stability and performance of an electrode. In the case of traditional electrolytes (1 M MPF_6 in EC : DEC (M = Li/Na) and 0.8 M KPF_6), for BTEG stability at 1 A g^{-1} , the capacity retention recorded was 88% (LIBs), 77% (SIBs), and only 33% (PIBs) after 200 cycles, as shown in Fig. S25.†

These values can be justified by a stepwise understanding of electrolyte composition (concentration, cation solvation, and the type of anion), electrode (alkali metal/BTEG), and the SEI.

Raman spectral analysis helps us to understand the role of the interaction of salts and solvents in electrolytes. The peak at 712 cm^{-1} was assigned to the C=O bending of free EC molecules in EC:DEC solvent (Fig. S26†).⁵⁶ On the addition of salts (MPF₆) in EC:DEC, a peak at 720 cm^{-1} emerges, associated with the solvation of M⁺, as shown in Fig. 6a.⁵⁷ The solvated peaks for K⁺ (KPF₆) are weaker in intensity than those of Li⁺ and Na⁺, implying more side reactions in the case of KPF₆ on continuous cycling.⁵⁸

To investigate the influence of electrolyte stability with alkali metals, long-term charge–discharge tests were performed on symmetric cells at a current density of $10\text{ }\mu\text{A cm}^{-2}$ (Fig. 6b). The overpotential of the plating/stripping profile after SEI stabilization (10 h) was in the increasing order Li (5 mV) < Na (14 mV) < K (42 mV). After 120 h, the overpotential for the K-symmetric cell exhibits a marked increase in comparison to Li and Na. The severe voltage fluctuations associated with KPF₆ are due to large volume changes with K plating/stripping, electrolyte degradation, or SEI breakage.⁵⁸ The voltage changes were further validated using EIS recorded over continuous cycling (Fig. S27a–ff†). A comparable trend was also observed in the stability analysis of BTEG in a half-cell configuration with a capacity retention of 88% (LIBs), 77% (SIBs), and only 33% (PIBs) at 1 A g^{-1} after 200 cycles (Fig. S25†). Consequently, analysing the composition of the SEI formed at the BTEG/electrolyte interface is crucial. The SEI composition was studied by *ex situ* XPS before (0 nm) and after etching (30 nm) of the BTEG electrode (Fig. S28–S40†), which was cycled at 25 mA g^{-1} for 10 cycles (SEI stabilised) (Fig. 6c). SEI comprises inorganic and organic components associated with salt and solvent degradation from electrolytes. Inorganic compounds are versatile metal salts such as MF, MPF, and MPOF, which arise from MPF₆ and are crucial for SEI stability.⁵⁹ Organic compounds are generally polyesters (RO-COOM) and polyethers (RO-M) that arise from the decomposition of EC and DEC. The structural stability of the SEI can diminish due to the higher solubility of organic components in the organic electrolyte. XPS analysis facilitated the acquisition of the integral areas corresponding to distinct peaks, which enabled the estimation of the normalized abundance of each constituent. After 10 cycles (SEI stabilized), the SEI surface (0

nm) of the BTEG electrode, which is in contact with electrolyte, shows the highest inorganic content for PIBs compared to LIBs and SIBs. This deduces that the surface chemistry of BTEG in PIBs is different from that in LIBs and SIBs (Fig. S39†). The highest inorganic component for PIBs is due to the higher dissolution of organic moieties in the organic electrolyte for PIBs than for LIBs and SIBs.^{60,61} The difference in solubility of the SEI can be understood by the variation in Lewis acidity of alkali ions. This leads to different solubilities of the electrolyte decomposition products (SEI), as well as variation in the reduction stability of EC:DEC solvent and hexafluorophosphate anions in the electrolyte. When the alkali cation size increases, going from Li⁺ to Na⁺, the SEI surface has been observed to be rich in inorganic content, due to high dissolution of organic species. Therefore, it is likely that the high inorganic content of the SEI surface in PIBs can be attributed to the higher dissolution of organic content at the BTEG/electrolyte interface. This suggests that the SEI formation mechanism is different for PIBs compared to LIBs and SIBs. Consequently, this infers that the higher solubility of organic components of the SEI is a contributing factor for diminished charge storage performance with KPF₆, which is 33% capacity retention (Fig. S25†) compared to LIBs (88%) and SIBs (77%) at 1 A g^{-1} after 200 cycles. Also, after etching (30 nm), the results indicate that the inorganic content is similar for LIBs, SIBs and PIBs (Fig. S38 and S40†). This suggests that there could be variation in reaction dynamics as the SEI grows influenced by alkali-ion–solvent complexes. It is noteworthy that the SEI from KPF₆ is dissolving and forming continually over cycles and irreversibly consuming electrolyte, thereby causing high-capacity fade within 200 cycles. Therefore, due to the unstable SEI in PIBs with KPF₆, selecting electrolytes that form a robust SEI is paramount.

In order to resolve the poor cycling stability issue of BTEG with KPF₆ for PIBs, we opted for HCE, *i.e.*, 5 M KFSI. BTEG with 5 M KFSI exhibits a significantly higher capacity retention of 91% at 1 A g^{-1} , compared to KPF₆ (33%) after 200 cycles (Fig. 7a). Raman analysis reveals that 5 M KFSI (HCE) shows stronger solvation than KPF₆ in EC:DEC, as shown in Fig. 7b and c. Generally, the types of solvation structures are solvent-separated ion pairs (SSIPs), contact ion pairs (CIPs), or aggregates (AGGs).⁶² KPF₆-based dilute electrolyte exhibits SSIPs, with peaks at 717 cm^{-1} (free EC molecule) and 744 cm^{-1} (K⁺-solvated EC), while in 5 M KFSI (HCE), the two peaks merge, indicating

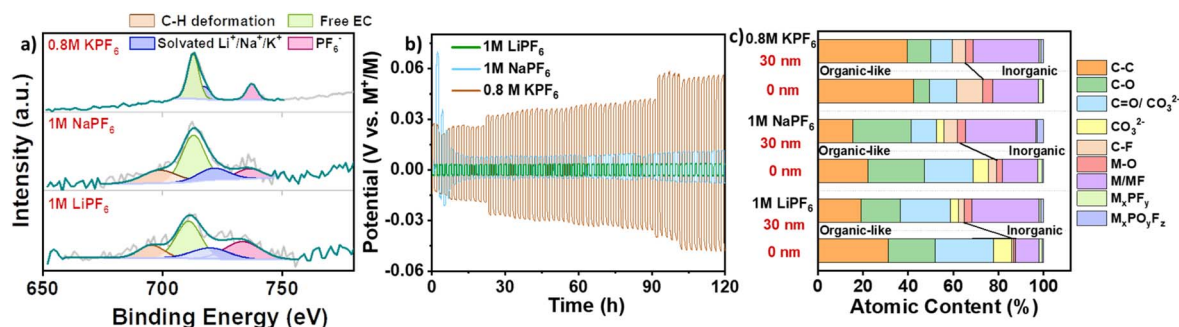


Fig. 6 (a) Raman spectra, (b) symmetric cell, and (c) *ex situ* XPS of BTEG compared for 1 M LiPF₆, 1 M NaPF₆, and 0.8 M KPF₆ in EC:DEC solvent.

the formation of CIPs (731 cm^{-1}) and AGGs (739 cm^{-1}). Additionally, the formation of CIPs and AGGs for KFSI suggests fewer free anions and higher solvation of K^+ in HCE, which helps reduce side reactions between K and electrolyte.

Fig. 7d compares the stability of the K/K symmetric cell with 5 M KFSI and 0.8 M KPF_6 (the zoomed in charge-discharge profile is shown in Fig. S41).[†] In the case of KFSI, the overpotential remains consistent throughout cycling, while KPF_6 shows a severe increase in overpotential. The increasing polarization losses in KPF_6 are confirmed by EIS (Fig. S42[†]), where the value of R_{CT} fluctuates (increases and decreases) with SEI dissolution and thickening through cycling. In contrast, for KFSI, R_{CT} tends to stay constant even after prolonged cycling (Fig. 7e). Due to the high stability with KFSI, capacity retention attained for BTEG electrodes is superior to that for KPF_6 -based cells.

The nature of SEI composition was further probed and compared for KFSI (HCE) and KPF_6 using XPS spectra (Fig. 7f–i). The KPF_6 -based SEI shows an organic-rich SEI layer due to the co-decomposition of both salts and solvents, as its solubility in EC:DEC is limited to a 0.8 M concentration (Fig. 7f). On the other hand, a high concentration of KFSI salt (5 M) on decomposition shows an inorganic-rich layer with KF, K_2CO_3 , K_2SO_4 , and K_2S components. The SEI layer comprises only 77% (before

etching) to 68% (after etching) reduction in the organic component with KPF_6 , while a considerable decrease from 65% (0 nm) to 34% (30 nm) with KFSI (Fig. 7f). Carbon and fluorine are the crucial elements that dictate SEI stability. Fig. 7g shows F 1s spectra which exhibit three peaks at around 683.3 eV (K–F), 688.1 eV (C–F), and 686.1 eV (K_xPF_y , K_xFSN) for KPF_6 and KFSI.⁶³ It is worth mentioning that the C–F peak disappears after etching (30 nm) for KFSI, while reduction is minimal for KPF_6 . It is to be noted that further fitting analysis reveals the presence of a B–F bond ($\sim 194\text{ eV}$)⁶⁴ which suggests that fluorine present in the electrolyte could interact with boron species (Fig. S44[†]). Fig. 7h shows C 1s spectra with a strong peak of the C–C bond at 284.8 eV, along with a K 2p peak, which is around 293 eV. It clearly shows a significant reduction of the carbon peak relative to K–F after etching for KFSI compared to KPF_6 . Additionally, a K–F component responsible for electron passivation in the SEI changes from 2.2% (0 nm) to 12.5% (30 nm) for KPF_6 and hugely for 3.06% (0 nm) to 20.5% (30 nm) for KFSI (HCE) (Fig. 7i). The inorganic components provide rigid and robust support on the anode surface.⁶⁵ High inorganic content and KF-rich species are attributed mainly to salt (anion) decomposition and scarcity of solvent molecules in HCE. This scarcity allows the salt anions into the first solvation shells, which lowers the LUMO energy level of FSI^- , leading to

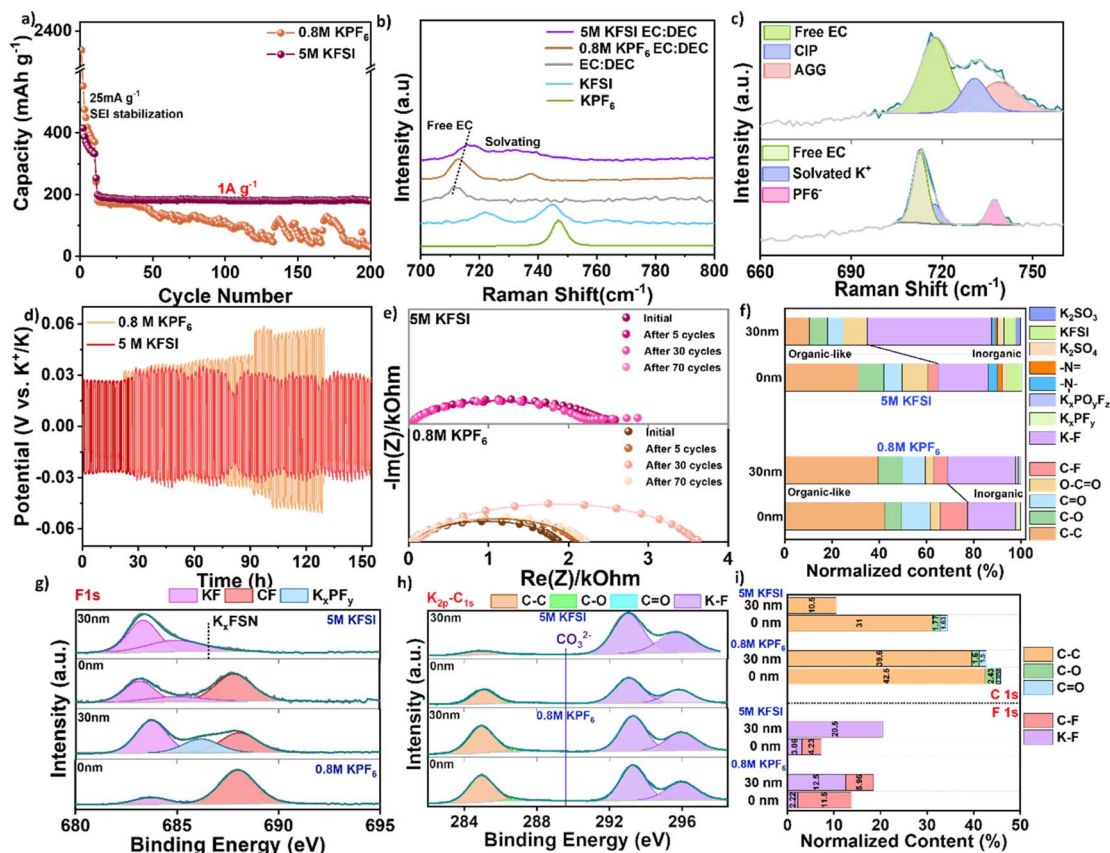


Fig. 7 Characterization comparing 0.8 M KPF_6 and 5 M KFSI for PIBs. (a) Cycle performance of BTEG at 25 mA g^{-1} and 1 A g^{-1} . (b) Comparison of Raman spectra of salts and carbonate solvent, and (c) deconvoluted spectra for electrolytes. (d) Voltage profiles for a K–K symmetric cell at $10\text{ }\mu\text{A cm}^{-2}$ and (e) Nyquist plots with impedance evolution for the symmetric cell. *Ex situ* XPS spectra comparing the SEI before and after etching on the BTEG anode, comparing (f) all elements (atomic%), (g) F 1s and (h) K 2p–C 1s, and (i) C 1s and F 1s (atomic%) with KPF_6 and 5 M KFSI.

preferential anionic decomposition.⁶⁶ Hence, the inorganic component with KF-rich moieties and high solvation helps attain highly stable BTEG (91% at 1 A g⁻¹ after 200 cycles) in PIBs. Thus, BTEG and symmetric K cells with 5 M KFSI-based electrolytes show exceptionally high stability compared to the KPF₆ electrolyte.

3. Conclusion

In conclusion, we have demonstrated that BTEG displayed an excellent performance characteristic for three alkali ion batteries. BTEG exhibits a reversible capacity of 1014 mA h g⁻¹, 295 mA h g⁻¹, and 369 mA h g⁻¹ at a current density of 25 mA g⁻¹ for LIBs, SIBs, and PIBs, respectively. At 1 A g⁻¹, BTEG exhibited a capacity retention of 75%, 100%, and 87% for LIBs, SIBs, and PIBs after 1000 cycles. Mechanistic and electrokinetic analysis showed charge storage characteristics, *i.e.*, intercalation-adsorption for LIBs and dominantly adsorption for SIBs and PIBs for BTEG, making BTEG a desirable candidate compared to graphite (only intercalation) for alkali-ion batteries. Investigations into electrolyte interactions with the anode material emphasized the importance of electrolyte/SEI engineering for PIBs. A highly concentrated electrolyte (5 M KFSI) markedly provides a robust SEI on the BTEG surface compared to the fragile SEI obtained from conventional low-concentration electrolytes such as 0.8 M KPF₆. The robustness can also be associated with high inorganic content for 5 M KFSI in the SEI, which helps attain long-term stability for up to 1000 cycles.

4. Experimental section

4.1 Materials

Graphite powder and sodium nitrate were supplied by Sigma-Aldrich. Concentrated sulfuric acid (95–95%), hydrogen peroxide (30%), and boric acid were acquired from Emparta, Merck. Potassium permanganate was acquired from Thomas Baker. Lithium hexafluorophosphate in ethylene dicarbonate, diethyl carbonate (EC:DEC), sodium hexafluorophosphate, potassium hexafluorophosphate, EC, and DEC battery grade were procured from Sigma-Aldrich. Potassium bis-fluorosulfonylimide was sourced from TCI Chemicals. A Whatman glass fiber separator was procured from Sigma-Aldrich.

4.2 Synthesis of BTEG

BTEG was prepared from thermal shock treatment of graphene oxide (GO). GO was synthesized by the modified Hummers' method.⁶⁷ GO was introduced into a preheated tube furnace at 350 °C in air for 5 minutes (termed TEG-350). 50 mg of thermally exfoliated graphene, TEG-350, was mixed with 50 mg of boric acid in 70 ml water to attain a homogeneous dispersion. It was completely dried at 70 °C overnight in an oil bath and boric acid with TEG-350 was ground using a mortar and pestle. Black-coloured powder was kept in a tube furnace in argon at full 0 °C for 3 h with a ramping rate of 5 °C min⁻¹. The pyrolysis product

with boron oxide residues from BTEG was washed with warm water (70 °C). The dispersion was centrifuged three times at 10 000 rpm for 5 min and finally washed with ethanol and dried overnight at 60 °C.

4.3 Control experiments

In order to understand the effect of boron doping, initial reversible capacity was recorded for TEG, which is prepared by annealing TEG-350 at 800 °C for 3 h with a ramping rate of 5 °C min⁻¹ without the boric acid precursor.

Data availability

The data supporting this article have been included as part of the ESI.†

Author contributions

Surishi Vashishth and Ujjwal Vidyarthi have contributed equally.

Conflicts of interest

The authors declare no competing financial interest.

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